OPLOGS-05: Querying & Parsing Logs

> **Series:** OPLOGS — OpenPipeline Logs | **Notebook:** 5 of 8 | **Created:** December 2025 | **Last Updated:** 10/14/2026

DQL Fundamentals and DPL Pattern Matching

This notebook covers DQL query syntax, filtering, string matching, and DPL (Dynatrace Pattern Language) for extracting structured data from logs.

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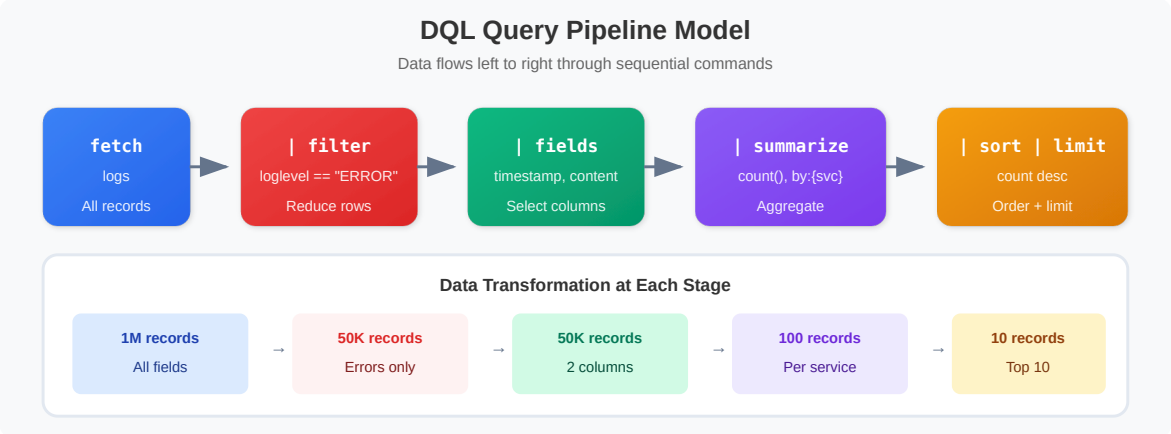
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Prerequisites

- ✅ Access to a Dynatrace environment with log data
- ✅ Completed OPLOGS-01 through OPLOGS-04
- ✅ Basic understanding of regular expressions (helpful)

1. DQL Fundamentals

Query Structure




```
fetch logs, from: now() - 1h          // Data source and time range
| filter &lt;condition&gt;                    // Filter records
| fields &lt;field1&gt;, &lt;field2&gt;          // Select fields
| sort &lt;field&gt; desc                   // Order results
| limit 100                           // Limit output
```

🚨 Critical DQL Rules (DQL ≠ SQL)

✅ DQL Syntax	❌ SQL Syntax
`filter status == "ERROR"`	`WHERE status = 'ERROR'`
`in(field, {"a", "b"})`	`field IN ('a', 'b')`
`summarize {count = count()}`	`SELECT COUNT(*)`
`isNull(field)`	`field IS NULL`

```
// Basic log query - explore recent logs
fetch logs, from: now() - 1h
| fields timestamp, content, loglevel, dt.entity.host
| limit 10
```

```
// Query with time range options
fetch logs, from: now() - 24h, to: now() - 12h
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

2. Filtering Logs

Comparison Operators

Operator	Description	Example
`==`	Equals	`loglevel == "ERROR"`
`!=`	Not equals	`status != "NONE"`
`>`, `>=`	Greater than	`count > 100`
`<`, `<=`	Less than	`duration < 1s`

Logical Operators

Operator	Description	Example
`AND`	Both conditions	`loglevel == "ERROR" AND isNotNull(host)`
`OR`	Either condition	`loglevel == "ERROR" OR loglevel == "WARN"`
`NOT`	Negate	`NOT contains(content, "health")`

> Compare durations with duration literals — `5s`, `100ms` — never a bare number: `duration < 1000` compares an integer with a duration and silently matches nothing.

```
// Filter by log level
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| fields timestamp, content, dt.entity.host
| limit 20
```

```
// Multiple conditions with AND/OR
fetch logs, from: now() - 1h
| filter (loglevel == "ERROR" OR loglevel == "WARN")
        AND isNotNull(dt.entity.host)
| summarize {count = count()}, by: {loglevel, dt.entity.host}
| sort count desc
| limit 15
```

```
// Filter using in() for multiple values
fetch logs, from: now() - 1h
| filter in(loglevel, {"ERROR", "WARN", "INFO"})
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

3. String Matching Functions

Function	Description	Example
`contains(str, substr)`	Substring match	`contains(content, "error")`
`startsWith(str, prefix)`	Prefix match	`startsWith(content, "[ERROR]")`
`endsWith(str, suffix)`	Suffix match	`endsWith(content, "failed")`
`matchesPhrase(str, phrase)`	Word boundary match	`matchesPhrase(content, "connection refused")`
`matchesValue(str, pattern)`	Exact or wildcard match	`matchesValue(host, "web-*")`

```
// Search for specific content patterns
fetch logs, from: now() - 1h
| filter contains(content, "Exception")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| fields timestamp, content_preview, dt.entity.host
| limit 15
```

```
// Use matchesPhrase for word-boundary matching
fetch logs, from: now() - 1h
| filter matchesPhrase(content, "connection")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 10
```

```
// Exclude patterns with NOT
// status == "ERROR" covers every error-class level (ERROR, SEVERE, CRITICAL,
FATAL, …);
// loglevel == "ERROR" alone misses SEVERE and the rest
fetch logs, from: now() - 1h
| filter status == "ERROR"
| filter NOT matchesPhrase(content, "health")
| filter NOT matchesPhrase(content, "heartbeat")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 10
```

4. Field Selection and Transformation

Field Commands

Command	Description
`fields`	Select specific fields only
`fieldsAdd`	Add computed fields
`fieldsRemove`	Remove fields

```
// Add computed fields
fetch logs, from: now() - 1h
| filter status == "ERROR"
| fieldsAdd severity = if(contains(content, "critical"), "CRITICAL",
                       else: if(contains(content, "fatal"), "FATAL",
                       else: loglevel))
| fieldsAdd content_length = stringLength(content)
| fields timestamp, severity, content_length, dt.entity.host
| limit 20
```

```
// String manipulation functions
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| fieldsAdd pod_short = substring(k8s.pod.name, from: 0, to: 30)
| fields timestamp, k8s.namespace.name, pod_short, loglevel
| limit 10
```

5. DPL (Dynatrace Pattern Language) Parsing

DPL extracts structured data from unstructured log content.

DPL Matchers

Matcher	Description	Matches
`INT`	Integer	`42`, `-17`
`DOUBLE`	Decimal	`3.14`, `-0.5`
`IPADDR`	IP address	`192.168.1.1`, `::1`
`LD`	Line data (to delimiter)	Any text
`WORD`	Word characters	`hello`, `user123`
`SPACE`	Whitespace	spaces, tabs
`JSON`	JSON structure	`{"key": "value"}`

Pattern Syntax

Syntax	Description
`MATCHER:fieldname`	Extract to named field
`MATCHER`	Match but don't extract
`MATCHER?`	Optional matcher
`'literal'`	Match exact string
`(opt1|opt2)`	Match alternatives

A `parse` pattern must match from the **start** of the string (trailing text is fine). To find a value anywhere, prefix `DATA?` before a literal, or use `parseAll()` for matcher-only patterns — see FAQ-15 §5.

```
// Parse log level from content (e.g., "[ERROR] message")
fetch logs, from: now() - 1h
| filter startsWith(content, "[")
| parse content, "'[' LD:parsed_level ']'"
// keep only supported loglevel values — bracketed timestamps and message
fragments are not levels
| fieldsAdd parsed_level = upper(parsed_level)
| filter in(parsed_level, {"EMERGENCY", "ALERT", "CRITICAL", "SEVERE",
"ERROR", "FATAL", "WARN", "NOTICE", "INFO", "DEBUG", "TRACE"})
| summarize {count = count()}, by: {parsed_level}
| sort count desc
| limit 10
```

```
// Parse HTTP-style logs
// Example: GET /api/users 200 45ms -&gt; GET, /api/users, 200
// Returns rows only for single-spaced lines of this exact shape; the range
filter drops counters in the status position
fetch logs, from: now() - 1h
| filter contains(content, "GET") OR contains(content, "POST")
| parse content, "[A-Z]+:method ' ' NSPACE:path ' ' INT:status_code ' '"
| filter status_code &gt;= 100 and status_code &lt;= 599
| summarize {count = count()}, by: {method, status_code}
| sort count desc
| limit 15
```

```
// Parse key-value pairs
// Example: "user=john action=login status=success"
// parseAll returns every user= value in the line (parse would return only
one);
// the !&amp;lt;&amp;lt; lookbehind stops "superuser=root" or "service_user=x"
matching as user=
fetch logs, from: now() - 1h
| fieldsAdd usernames = parseAll(content, "!&amp;lt;&amp;lt;[A-Za-z0-9_] 'user=' [A-
Za-z0-9._@-]+:username")
| filter arraySize(usernames) &amp;gt; 0
| expand username = usernames
| summarize {count = count()}, by: {username}
| sort count desc
| limit 10
```

```
// Parse IP addresses from logs
fetch logs, from: now() - 1h
// parseAll finds every IP anywhere in the line; parse would only match one at
the very start
// IPADDR also matches IPv4-shaped version strings (rv:1.9.2.20) and IPv6
fragments (::0000);
// OPLOGS-08 §3 shows a bounded IPv4 pattern for classification
| fieldsAdd client_ip = parseAll(content, "IPADDR:ip")
| filter arraySize(client_ip) &amp;gt; 0
| expand client_ip
| summarize {count = count()}, by: {client_ip}
| sort count desc
| limit 15
```

6. Working with JSON Logs

Many applications emit JSON-formatted logs. DQL provides tools to work with JSON content.

```
// Find JSON-formatted logs
fetch logs, from: now() - 1h
| filter startsWith(content, "{")
| fieldsAdd content_preview = substring(content, from: 0, to: 150)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 10
```

```
// Parse JSON and extract fields
fetch logs, from: now() - 1h
| filter startsWith(content, "{")
| parse content, "JSON:json_data"
| filter isNotNull(json_data)
| fields timestamp, json_data
| limit 10
```

```
// Access nested JSON fields
fetch logs, from: now() - 1h
| filter startsWith(content, "{")
| parse content, "JSON:json_data"
| filter isNotNull(json_data)
| fieldsAdd error_type = json_data[errorType]
| fieldsAdd status_val = json_data[status]
| filter isNotNull(status_val)
| summarize {count = count()}, by: {status_val, error_type}
| sort count desc
| limit 15
```

7. Null Handling

DQL uses three-valued logic. `NULL` comparisons require special functions.

❌ Wrong	✅ Correct
`field == null`	`isNull(field)`
`field != null`	`isNotNull(field)`

```
// Check for null values
fetch logs, from: now() - 1h
| summarize {
    total = count(),
    with_host = countIf(isNotNull(dt.entity.host)),
    without_host = countIf(isNull(dt.entity.host)),
    with_namespace = countIf(isNotNull(k8s.namespace.name)),
    without_namespace = countIf(isNull(k8s.namespace.name))
  }
```

```
// Use coalesce for default values
// coalesce returns the first non-null argument: the workload name, else the namespace, else "UNKNOWN".
// (coalesce(loglevel, …) would never fall back: loglevel is never null, and missing levels arrive as "NONE".)
fetch logs, from: now() - 1h
| fieldsAdd owner = coalesce(k8s.workload.name, k8s.namespace.name, "UNKNOWN")
| summarize {count = count()}, by: {owner}
| sort count desc
| limit 15
```

8. Advanced Parsing Examples

```
// Parse exception patterns
// Split the record into whitespace-separated tokens (splitByPattern with SPACE also splits on line breaks) and keep the ones that end in "Exception".
// A pattern like "LD:exception_type 'Exception'" anchors at the start of the line, so it
// captures the whole prefix ("2026-10-02 ERROR c.e.Svc - java.lang.NullPointer") instead
// of the class name; DPL does not backtrack, so a greedy word matcher cannot stop before
// the literal "Exception" either.
fetch logs, from: now() - 24h
| filter contains(content, "Exception")
| fieldsAdd token = splitByPattern(content, "SPACE")
| expand token
| filter endsWith(token, "Exception") OR endsWith(token, "Exception:")
| fieldsAdd exception_type = replaceString(token, ":", "")
| summarize {count = count()}, by: {exception_type}
| sort count desc
| limit 15
```

```
// Parse with optional fields
// Matches "error code=123" or "... error code=123 message=failed" anywhere in
the line
// (DATA? lets the pattern start mid-line; parse otherwise anchors at the
start)
fetch logs, from: now() - 1h
| filter contains(content, "error")
| parse content, "DATA? 'error' SPACE 'code=' INT:error_code (SPACE 'message='
LD:error_msg)?"
| filter isNotNull(error_code)
| summarize {count = count()}, by: {error_code}
| sort count desc
| limit 10
```

```
// Parse alternative formats
// Matches "user=john" or "username=john" anywhere in the line, every
occurrence (parseAll);
// the character class stops at the first character outside it, so
"user=jdoe," yields "jdoe"
// (a trailing LD would take the rest of the line), and the !&amp;lt;&amp;lt;
lookbehind skips "superuser=root"
fetch logs, from: now() - 1h
| fieldsAdd user_values = parseAll(content, "!&amp;lt;&amp;lt;[A-Za-z0-9_]
('user='|'username=') [A-Za-z0-9._@-]+:user_value")
| filter arraySize(user_values) &amp;gt; 0
| expand user_value = user_values
| summarize {count = count()}, by: {user_value}
| sort count desc
| limit 10
```

📝 Summary

In this notebook, you learned:

✅ **DQL fundamentals** - Query structure and syntax rules
✅ **Filtering** - Comparison operators, logical operators, in()
✅ **String matching** - contains, matchesPhrase, startsWith
✅ **Field manipulation** - fieldsAdd, computed fields
✅ **DPL parsing** - Matchers (INT, LD, IPADDR, JSON)
✅ **JSON handling** - Parsing and accessing nested fields
✅ **Null handling** - isNull, isNotNull, coalesce

➡️ Next Steps

Continue to **OPLOGS-06: Topology & Entity Context** to learn about entity context and relationships.

📚 References

- Dynatrace Query Language (DT docs)
- DQL functions (DT docs)
- Dynatrace Pattern Language (DT docs)
- DPL Architect (DT docs)

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